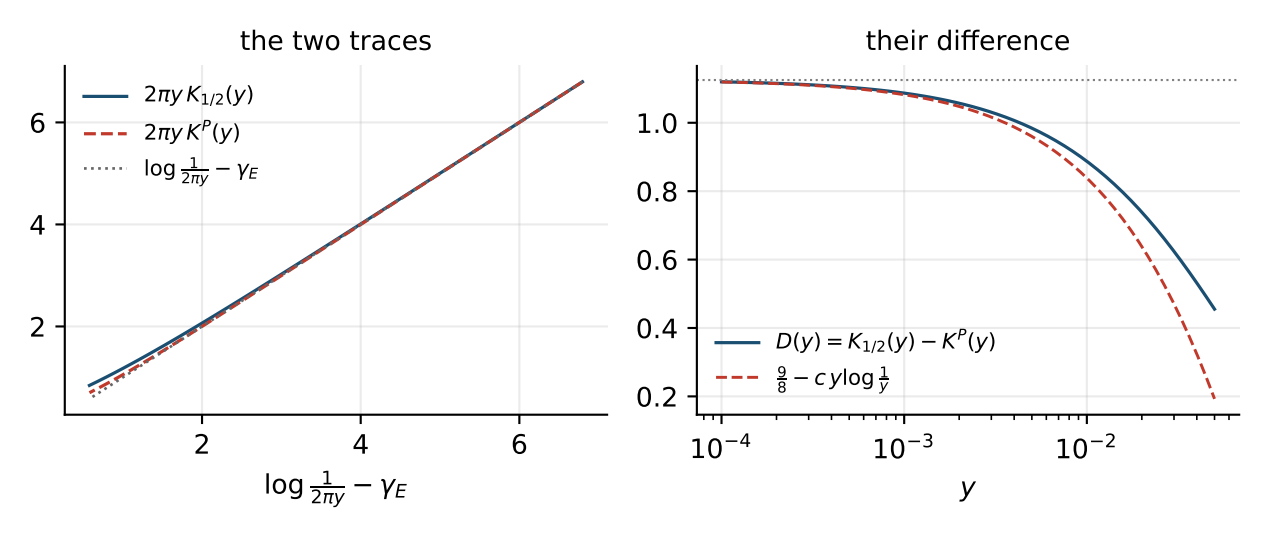
the two traces
$2\pi y\, K_{1/2}(y)$
$2\pi y\, K^P(y)$
$\log \frac{1}{2\pi y} - \gamma_E$
6
4
2
2
4
6
$\log \frac{1}{2\pi y} - \gamma_E$
their difference
1.0
0.8
0.6
0.4
0.2
$D(y) = K_{1/2}(y) - K^P(y)$
$\frac{9}{8} - c\, y \log \frac{1}{y}$
$10^{-4}$
$10^{-3}$
$10^{-2}$
$y$